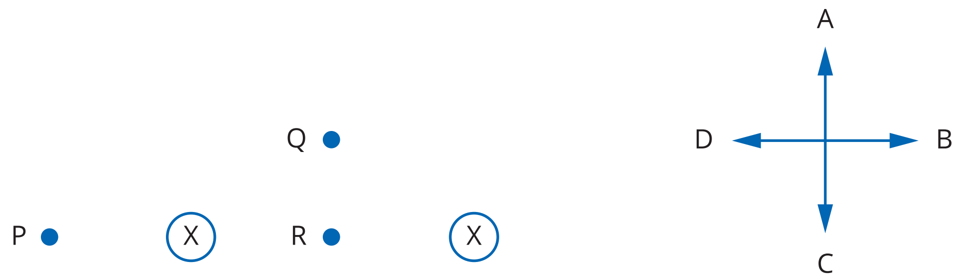
ATAR course **Year 12 Physics**

Practice Exam 2 (Unit 3 & 4) **Answers**

Section one: **Short answer** (52 marks)

Question 1 **(3 marks)**

The diagram represents two conductors, both perpendicular to the page and both carrying equal currents into the page. In these questions ignore any contribution from the Earth's magnetic field. The arrows A to D and letters E to G represent options from which you are to choose in answering the following questions.



E Out of page

F Intro page

G Zero field

What is the direction of the magnetic field due to the two currents at the following points?

a point P

b point Q

c point R

Description	Marks
a A	1
b C	1
c G	1
Total	3

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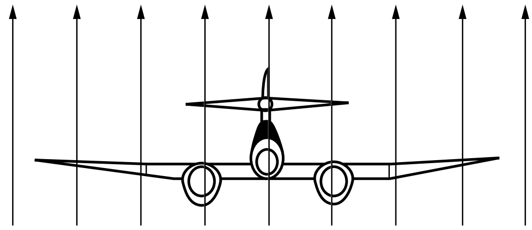
Question 2 **(4 marks)**

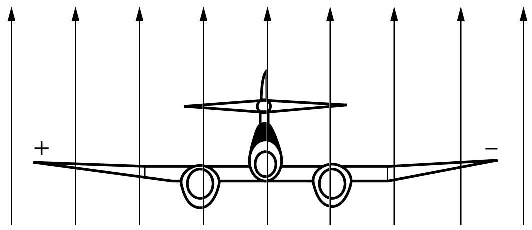
The figure below shows an aircraft flying at right angles to the Earth's magnetic field ($B = 2.0 \times 10^{-6}$ T) at a speed of 160 m s^{-1}. The length of the wings from tip to tip is 30 m.



a Determine the magnitude of the EMF generated across the tips of the aircraft's wings. (2 marks)

Description	Marks
EMF = $\ell vB = 30 \times 160 \times 2 \times 10^{-6}$	1
EMF = 9.60×10^{-3} V	1
Total	**2**

b On the diagram above, place a '+' sign at the end of the wing that will acquire a positive charge and a '–' sign at the end of the wing that will acquire a negative charge. Explain your answer in the space provided. (2 marks)



Description	Marks
Charges labelled correctly	1
Using the right-hand rule, the direction of force on positive charges will be to the left.	1
Total	**2**

Question 3 **(5 marks)**

To determine the electrical permittivity, ε, in a newly developed material, a student measured the force of repulsion between two charged spheres separated by 1.0 cm. When each sphere carried a charge of 1.0 μC, the force measured was 14.8 N.

a What is the electrical permittivity of this newly developed material? (3 marks)

Description	Marks
$F = \frac{1}{4\pi\varepsilon} \times \frac{q_1 q_2}{r^2}$ $\varepsilon = \frac{1}{4\pi F} \times \frac{q_1 q_2}{r^2}$	1
$\varepsilon = \frac{1}{4\pi \times 14.8} \times \frac{(1.0 \times 10^{-6})^2}{(0.01)^2}$	1
$\varepsilon = 5.38 \times 10^{-11}\ F\,\mathrm{m}^{-1}$	1
Total	**3**

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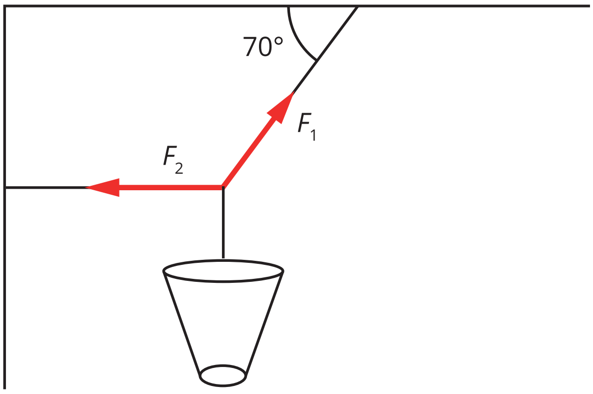
b Show how this permittivity compares to the permittivity of a vacuum. (2 marks)

Description	Marks
$\frac{\varepsilon}{\varepsilon_0} = \frac{5.38 \times 10^{-11}}{8.85 \times 10^{-12}}$	1
That is, $\varepsilon = \varepsilon_0 \times 6.08$	1
Total	2

Question 4 **(4 marks)**

Determine the tension in the two cords used to secure the 75 kg pot plant as represented in the diagram below.



Description	Marks
Recognition that pot is in static equilibrium, so $\Sigma F = 0$ Thus, upward component of F_1 = weight force of pot plant and F_2 = horizontal component of F_1	1
$F_1 \sin 70° = mg$ $F_1 = \frac{75 \times 9.8}{\sin 70°}$	1
$F_1 = 782$ N	1
$F_2 = F_1 \cos 70° = 268$ N	1
Total	4

Question 5 **(4 marks)**

A domestic microwave oven is rated at 1000 W and uses a frequency of 2500 MHz. Calculate the number of photons emitted per second by the magnetron that produces the microwaves in the oven.

Description	Marks
$E = hf = 6.63 \times 10^{-34} \times 2500 \times 10^{6} = 1.66 \times 10^{-24}$ J	1
$P = \frac{E}{t}$ Thus, $E = P \times t$	1
$E = 1000 \times 1 = 1000$ J s^{-1}	1
Number of photons per second = $\frac{1000}{1.66 \times 10^{-24}} = 6.02 \times 10^{26}$	1
Total	4

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Question 6 **(5 marks)**

A remote vehicle sits on an asteroid of diameter 28 km. On top of the remote vehicle is an object, at a height of 1.2 m from the surface of the asteroid. The remote vehicle propels the object vertically upwards at 0.5 m s^{-1}. The object returns to the asteroid's surface 208 seconds after launch.

a What is the acceleration due to gravity on the surface of the asteroid? (3 marks)

Description	Marks
$s = ut + \frac{1}{2}at^2$ $1.2 = -0.5 \times 208 + \frac{1}{2} \times g \times 208^2$	1
$g = \frac{(1.2 + 104) \times 2}{208^2}$	1
$g = 0.004\,86$ m s^{-2}	1
Total	**3**

b What is the mass of the asteroid? (2 marks)

Description	Marks
$g = \frac{Gm}{r^2}$ Thus, $m = \frac{gr^2}{G}$	1
$m = \frac{0.004\,86 \times (140\,00)^2}{6.67 \times 10^{-11}} = 1.43 \times 10^{16}$ kg	1
Total	**2**

Question 7 **(4 marks)**

An asteroid orbits the sun with a period of approximately 410 days. What is the average distance of the asteroid from the sun?

Description	Marks
$\frac{T^2}{r^3} = \frac{4\pi^2}{Gm_{Sun}}$ Thus, $r^3 = \frac{Gm_{Sun}T^2}{4\pi^2}$	1
$r^3 = \frac{6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times (410 \times 24 \times 3600)^2}{4\pi^2}$ $= 4.22 \times 10^{33}$	2
$r = 1.62 \times 10^{11}$ m	1
Total	**4**

Question 8 **(4 marks)**

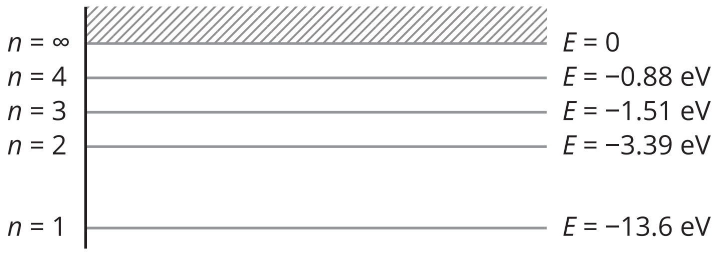
An alternating generator with a rotor turning at a frequency of 50 Hz has 200 turns in its stator armature coils. Each coil has an area of 100 cm². The magnetic field strength produced by the alternator's magnetising current is 0.50 T.

Determine the peak voltage output of the generator.

Description	Marks
$1\ \text{cm}^2 = 10^{-4}\ \text{m}^2$ So, $100\ \text{cm}^2 = 0.01\ \text{m}^2$	1
$EMF_{max} = -2\pi NBAf$	1
$EMF_{max} = -2\pi \times 200 \times 0.5 \times 0.01 \times 50$	1
$EMF_{max} = -314$ V	1
Total	**4**

Question 9 **(4 marks)**

Consider the energy level diagram for the hydrogen atom shown below. A photon of energy 14.0 eV collided with a hydrogen atom in the ground state.



a Explain why this collision will eject an electron from the atom. (2 marks)

Description	Marks
The single H electron is in the ground state, so it has energy of 13.6 eV.	1
The photon energy is greater than 13.6 eV and the energy of $n = 1$, so the electron will acquire sufficient energy to be ejected from the atom.	1
Total	**2**

b Calculate the energy of the ejected electron in electron volts and joules. (2 marks)

Description	Marks
Energy of ejected electron = 14.0 − 13.6 = 0.4 eV	1
In joules, $E = 0.4 \times 1.6 \times 10^{-19} = 6.40 \times 10^{-20}$ J	1
Total	**2**

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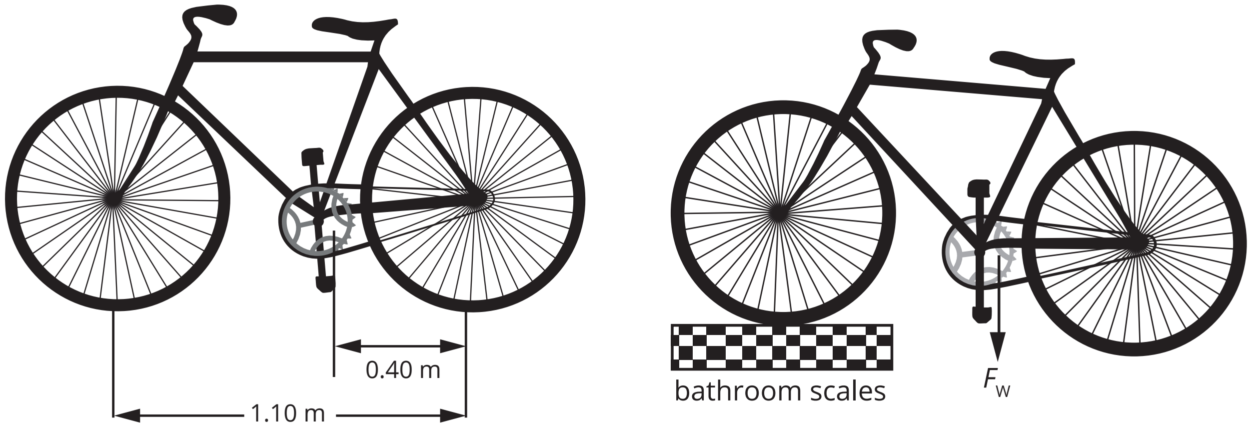
Question 10 **(4 marks)**

Rodney wants to determine the mass of his bicycle but the only equipment he has to weigh objects with is a set of bathroom scales. He has been able to determine that the centre of mass of the bicycle is 40 cm from the axis of the back wheel, and the distance between the wheel axles is 1.10 m.

Explain how Rodney can find the mass of his bike using only the bathroom scales. You may use a diagram to assist your explanation and indicate any mathematical relationships used to determine the bike's mass.



Description	Marks
Place the front wheel on the scales and take the reading on the scales	1
Take the movements about the centre of the back wheel. Σ clockwise moments = Σ anticlockwise moments	1
Let m_s = mass reading from bathroom scales. $m_s \times 9.8 \times 1.10 = F_w(1.10 - 0.40)$	1
Bike's mass = $\frac{F_w}{9.8}$	1
Total	**4**

Question 11 **(3 marks)**

Before the development of the Standard Model, it was thought that forces were exerted on particles by fields. In the Standard Model, it is believed that forces are exerted by the exchange of other particles.

a What name is given to the particles that give rise to forces? (1 mark)

Description	Marks
(Gauge) bosons	1
Total	**1**

b When two electrons approach each other, each experiences a force of repulsion from the other. How does the Standard Model explain this repulsion? (2 marks)

Description	Marks
The electrons exchange gauge bosons between them.	1
Each electron emits a photon which is exchanged, and this is what causes the repulsion between the two electrons.	1
Total	**2**

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Question 12 **(4 marks)**

The rest length of a spacecraft is 75 m. As it passes by a space station, its length is measured to be 60 m. What is the speed of the spacecraft relative to an observer on the space station?

Description	Marks
$\ell = \ell_0 \sqrt{1 - \frac{v^2}{c^2}}$ $v^2 = (1 - \frac{\ell^2}{\ell_0^{\,2}})c^2$	1
$v = \sqrt{(1 - \frac{\ell^2}{\ell_0^{\,2}})c^2} = \sqrt{(1 - \frac{60^2}{75^2})c^2}$	2
$v = 0.6c$	1
Total	**4**

Question 13 **(4 marks)**

The absorption spectra for distant stars are red shifted. When this was first observed, it was interpreted as a Doppler shift.

a Explain what is meant by a Doppler shift in the context of a red shift of light. (2 marks)

Description	Marks
Light is moving away from an observer.	1
As it moves away, its wavelength appears to lengthen.	1
Total	**2**

It was later realised that the red shift of light from distant stars provides evidence to support the idea that the universe is expanding.

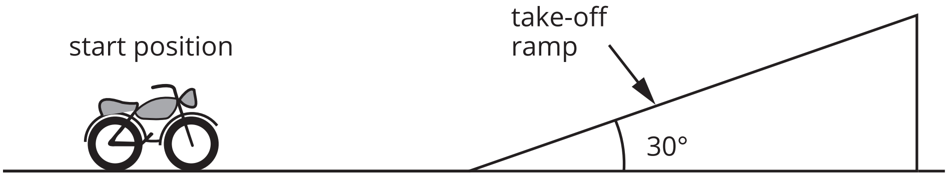
b Explain how the expansion of the universe accounts for the red shift of light from distant stars. (2 marks)

Description	Marks
Recognition that space itself is getting bigger	1
Recognition that the EMR waves ‘stretch’ with space and so their wavelength increases	1
Total	**2**

Section two: **Problem-solving** (91 marks)

Question 14 **(14 marks)**

A motorcycle stunt rider is designing a stunt to jump a series of trucks lined up side-by-side to give a total length of 50 m. The rider will need to reach a maximum height of 8 m to ensure that the trucks are cleared. The rider will take off from a ramp at an angle of 30°. The diagram below illustrates the jump.



a Calculate the speed at which the motorcycle needs to leave the ramp to reach the required height. (5 marks)

Description	Marks
$v^2 = u^2 + 2as$ $0 = u^2 + 2 \times -9.8 \times 8$	1
$u = \sqrt{16 \times 9.8} = \sqrt{156.8}$	1
$u = 12.5 \text{ m s}^{-1}$ That is, the initial velocity in the vertical direction is 12.5 m s^{-1} upwards.	1
Velocity off the 30° ramp: $v = \dfrac{12.5}{\sin 30°}$	1
$v = 25.0 \text{ m s}^{-1}$	1
Total	**5**

b What is this speed in kilometres per hour? (1 mark)

Description	Marks
$v = 25 \times 10^{-3} \times 3600 = 90.2 \text{ km h}^{-1}$	1
Total	**1**

c Confirm that this take-off speed is sufficient to cover the required distance. (5 marks)

Description	Marks
Time of flight: $v_v + u_v + at$ $-12.5 = 12.5 + (-9.8t)$	2
$t = \dfrac{25}{9.8} = 2.56 \text{ s}$	1
$v_h = 25.0 \cos 30° = 21.7 \text{ m s}^{-1}$	1
$s_h = v_h \times t = 21.7 \times 2.56 = 55.5 \text{ m}$	1
Total	**5**

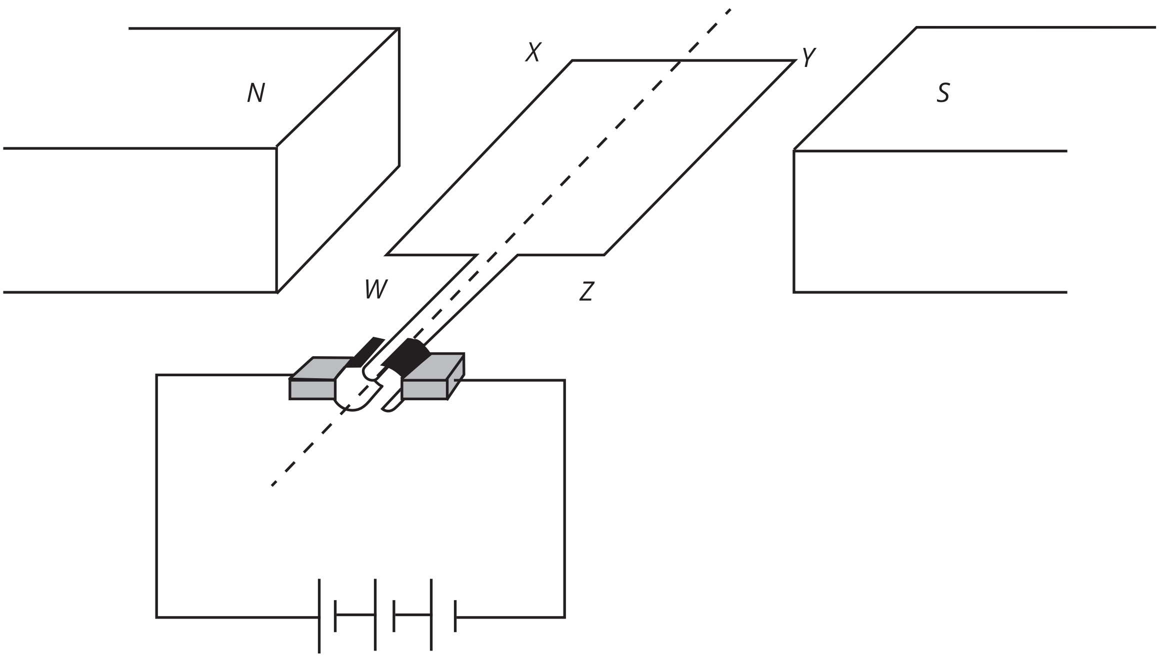
d The stunt rider plans to start the ride on the flat and accelerate towards the take-off ramp. Without changing the motorcycle or the take-off ramp, suggest what change to the design of the stunt can help ensure the rider takes off at the required speed. Explain your answer using the appropriate physics. (3 marks)

Description	Marks
Start the ride from a raised ramp.	1
As the rider goes down the ramp, acceleration due to gravity will help the rider gain velocity and reach the take-off ramp at a higher speed.	1
This is due to the conversion of potential energy that the rider has at the top of a ramp to kinetic energy as she/he goes down the ramp.	1
Total	**3**

Question 15 **(9 marks)**

The figure below shows a simplified diagram of a DC motor made by a physics student. It was built using two permanent magnets, a 6.0 V battery, a split ring commutator and a rectangular coil of wire consisting of 50 loops. As shown in the diagram, the coil is parallel to the magnetic field.

The resistance of the coil is 4.0 Ω. The length of side WX is 6.0 cm and the length of side XY is 4.0 cm. The magnetic field is 0.08 T.



a Calculate the current flowing through the coil of wire. (2 marks)

Description	Marks
$V = IR$ $I = \frac{V}{R} = \frac{6}{4}$	1
$I = 1.5\ \text{A}$	1
Total	**2**

b Determine the size of the force acting on side WX when the coil is in the position shown in the diagram. (2 marks)

Description	Marks
$F = nI\ell B \sin 90° = 50 \times 1.5 \times 0.04 \times 0.08 \times 1$	1
$F = 0.024$ N (apply follow-through marks for value of current from part a)	1
Total	**2**

c What is the size of the force acting on side XY when the coil is in the position shown in the diagram? Support your answer with suitable reasoning. (2 marks)

Description	Marks
$F = 0$ N	1
$F = nI\ell B \sin 0° = 50 \times 1.5 \times 0.04 \times 0.08 \times 0$ OR recognition that a conductor parallel to a magnetic field does not experience a force	1
Total	**2**

d State the role of the split ring commutator in this circuit and explain how it achieves its purpose. (3 marks)

Description	Marks
When the loop is at 90° to the magnetic field, the split in the commutator stops current flowing through the loop. Its momentum carries it past the vertical.	1
This reverses the direction of the current in the loop every half turn.	1
This reverses the direction of the forces acting on the sides WX and YZ, ensuring the continuous rotation of the loop.	1
Total	**3**

Question 16 **(16 marks)**

Electricity from a water-driven alternator of output 1000 V AC is transmitted by several kilometres of transmission lines to a step-down transformer which provides 250 V AC for a mill.

a Briefly describe the principle by which an alternator/generator produces electricity. Support your answer with a diagram. (4 marks)

Description	Marks
A coil of wire rotates in a magnetic field.	1
A current is created in the conductor as it rotates.	1
Diagram shows: • magnet surrounding a coil • coil rotating in magnetic field	 1 1
Total	**4**

b The primary (high-voltage) winding of the transformer consists of 2000 turns. How many turns are there in the secondary winding? (2 marks)

Description	Marks
$\frac{N_p}{N_s} = \frac{V_p}{V_s}$ $N_s = \frac{V_s}{V_p} \times N_p$	1
$N_s = \frac{250}{1000} \times 2000 = 500$ turns	1
Total	**2**

The alternator produces 200 kW of electrical power when the mill is working. The total resistance of the high-voltage transmission lines is 0.2 Ω.

c What is the current flowing in the transmission lines when the mill is working? (3 marks)

Description	Marks
$P = VI$ $I = \frac{P}{V}$	1
$I = \frac{200\,000}{1000}$	1
$I = 200$ A	1
Total	**3**

d What is the power loss in the high-voltage transmission lines when the mill is working? (3 marks)

Description	Marks
$P = I^2R$	1
$P = (200)^2 \times 0.2$	1
$P = 8000$ W	1
Total	**3**

e The mill engineer is concerned about the power loss in the transmission lines from the alternator to the mill. State what steps the engineer could take to reduce this loss. Explain how this can reduce the transmission losses. (4 marks)

Description	Marks
Power from the alternator can be stepped up before it is transmitted.	1
This will send the power at high voltage but low current—this follows from $P = VI$.	2
Thus, power lost in the transmission lines will be reduced due to $P = I^2R$.	1
Total	**4**

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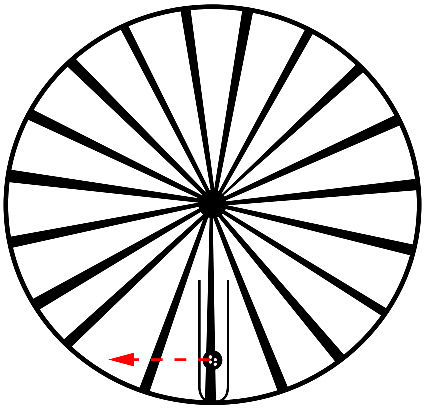
Question 17 **(12 marks)**

A centrifuge is a device that rotates test tubes at very high speeds to separate components suspended in a liquid. The diagram below represents a centrifuge containing a test tube.



The centrifuge rotates at 75 000 rpm (revolutions per minute) in a clockwise direction. The open end of the test tube is 5 cm from the axis of rotation and the closed end is 15 cm from the axis.

a At any point in its motion, a suspended particle has the tendency to move in a straight line as represented in the diagram above. Explain why it does not continue to move in a straight line but rotates around the axis of the centrifuge. (2 marks)

Description	Marks
The base of the test tube exerts a centripetal force which is transferred through the fluid to particles suspended in the liquid.	1
The direction of this force is towards the centre of the centrifuge which causes it to travel in a circular path.	1
Total	**2**

b Determine the centripetal acceleration in terms of '*g*' (acceleration due to gravity) at the open end of the test tube. (6 marks)

Description	Marks
$T = \frac{60}{75\,000}$	1
$v = \frac{2\pi r}{T} = \frac{2\pi \times 0.05 \times 750\,000}{60}$	1
$v = 392\ \text{m s}^{-1}$	1
$a = \frac{V^2}{r} = \frac{392^2}{0.05}$	1
$a = 3.08 \times 10^6\ \text{m s}^{-2}$	1
In terms of '*g*': $a = \frac{3.08 \times 10^6}{9.8} = 3.14 \times 10^5$	1
Total	**6**

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c What is the centripetal acceleration at the closed end of the test tube? (2 marks)

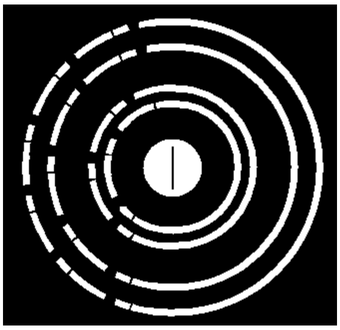
Description	Marks
Recognition that the radius to the closed end is three times the radius of the open end	1
$a = 3 \times 3.08 \times 10^6 = 9.24 \times 10^6$ m s^{-2} (if expressed in terms of 'g': $a = 9.43 \times 10^6$)	1
Total	2

d If the contents of the test tube have a mass of 11.5 g, what force must the bottom of the test tube withstand? (2 marks)

Description	Marks
$F = ma = 0.0115 \times 9.24 \times 10^6$	1
$F = 1.06 \times 10^5$ N	1
Total	**2**

Question 18 **(12 marks)**

The figure below shows a photograph of the diffraction patterns obtained by a group of students. They investigated the diffraction pattern created when a beam of electrons passed through a very thin film of gold foil and then the pattern created when X-rays were used instead of electrons.



The electrons used in the experiment were accelerated to a speed of 1.8×10^7 m s^{-1} before passing through the gold foil.

a Calculate the de Broglie wavelength of the electrons used. Show your working. (2 marks)

Description	Marks
$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 1.8 \times 10^7}$	1
$\lambda = 4.04 \times 10^{-11}$ m	1
Total	**2**

b State and explain the effect on the electron diffraction pattern if the accelerating voltage applied to the electron were increased. (3 marks)

Description	Marks
The radius of the diffraction pattern would decrease.	1
As $\lambda = \frac{h}{mv}$, an increase in velocity will reduce the wavelength.	1
As diffraction is proportional to the wavelength, decreasing λ will result in the radius of the diffraction pattern getting smaller.	1
Total	**3**

c Determine the energy of a photon of the X-rays used. Show your working. (3 marks)

Description	Marks
Recognition that since the diffraction pattern for the X-rays and electron are the same, the wavelength of the X-rays will be the same as the de Broglie wavelength of the electrons	1
$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{4.04 \times 10^{-11}}$	1
$E = 4.92 \times 10^{-15}$ J	1
Total	**3**

d State why the diffraction pattern obtained using electrons is the same as that obtained using X-rays. (2 marks)

Description	Marks
Diffraction is a property of wave-like behaviour.	1
Thus, as the electrons have exhibited a diffraction pattern, they must have wave-like properties.	1
Total	**2**

e Observations of the diffraction patterns for the interaction of X-rays and other electromagnetic radiation, with matter or through double slits (as in Young's experiment), support the wave model of light. However, we also observe properties of light that suggest it has a particle-like nature. Give two pieces of evidence that support the particle model of light. (2 marks)

Description	Marks
Any two appropriate pieces of evidence such as: • photoelectric effect • black body radiation • quantisation of light energy as given by $E = hf$	2
Total	**2**

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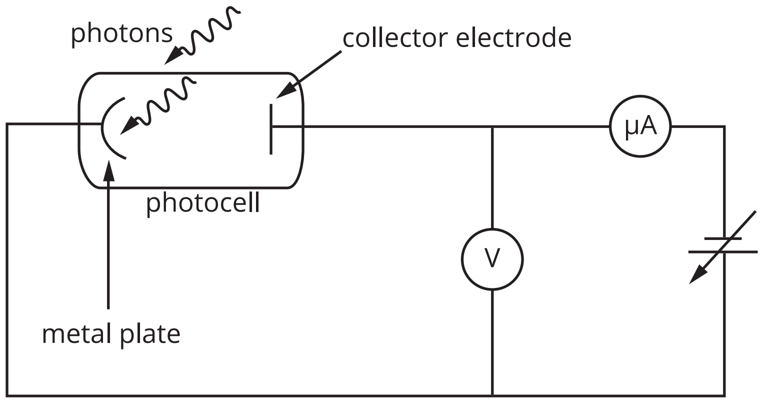
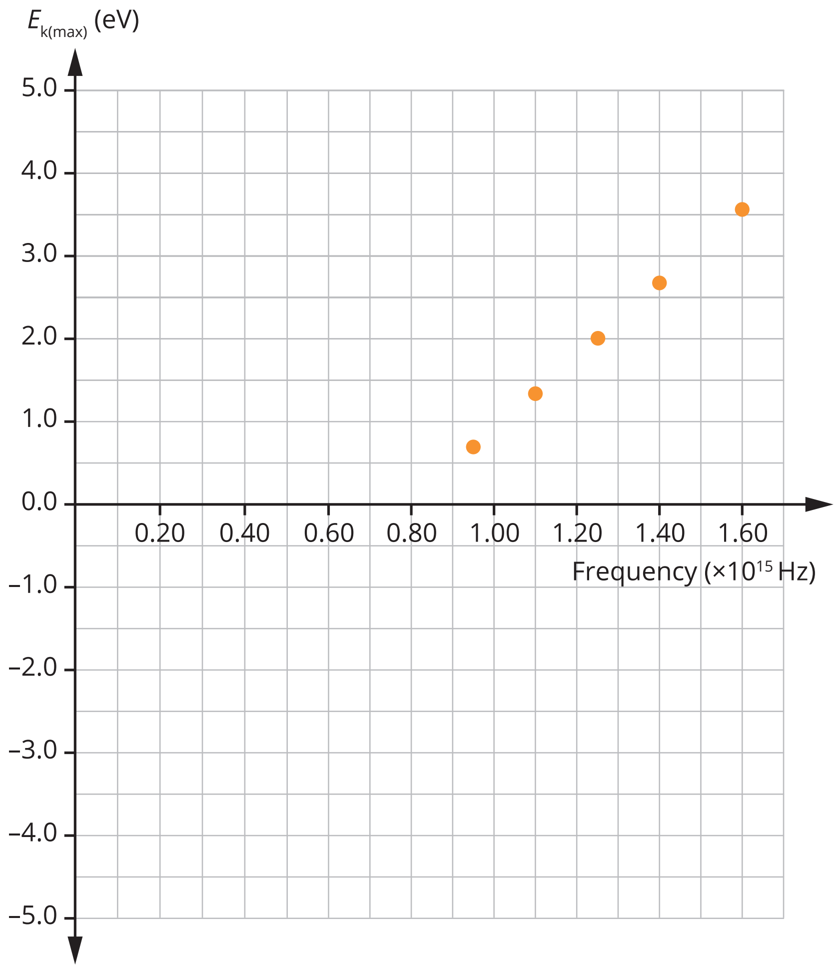
Question 19 **(16 marks)**

In an experiment designed to investigate the photoelectric effect, a group of physics students allowed light of various frequencies to fall on a metal plate coated in scandium (Sc) inside a photocell. The figure below shows the electric circuit used by the students to collect their data.



The data they collected is shown in the graph below.



a From the data in the graph, what value would the students obtain for Planck's constant? Include the units in your answer. (4 marks)

Description	Marks
Drawing of line of best fit	1
h = gradient of line of best fit = $\frac{3.5 - 0}{(1.6 - 0.8) \times 10^{15}}$	1
$h = 4.38 \times 10^{-15}$ eV s (or 7×10^{-34} J s)	2
Total	**4**

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b From the data on the graph, what is the minimum energy required to remove photoelectrons from the scandium (Sc) plate? (2 marks)

Description	Marks
Drawing of line of best fit back to intercept energy axis	1
3.5 eV	1
Total	**2**

c Explain why the existence of a threshold frequency supports the particle model and not the wave model for light. (2 marks)

Description	Marks
According to the wave model, a threshold frequency for the incident photons would not be required. This is because the wave model predicts that all the incident photons would be absorbed, resulting in the electrons eventually gaining enough energy to escape the metal. This behaviour is not observed.	1
The particle model correctly predicts that unless an incident photon has sufficient energy to enable an electron to escape the metal, the photon will not be absorbed. The threshold frequency is the minimum frequency for which photons will have enough energy to liberate an electron.	1
Total	**2**

The students replaced their scandium (Sc) photocell with one in which the metal plate was coated in caesium (Cs). Caesium has a work function of 2.1 eV.

The students then shone light from a 5.0 mW (5.0×10^{-3} W) monochromatic blue laser of wavelength λ = 460 nm onto the caesium photocell.

d Determine the velocity of the electrons ejected from the caesium. (6 marks)

Description	Marks
$E = \frac{hc}{\lambda} = \frac{4.38 \times 10^{-15} \times 3 \times 10^{8}}{460 \times 10^{-9}}$	1
E = 2.86 eV	1
$E_k = 2.86 - 2.1 = 0.76$ eV	1
Thus, $E_k = 1.22 \times 10^{-19}$ J	1
$E_k = \frac{1}{2}mv^2$ Thus, $v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 1.22 \times 10^{-19}}{9.11 \times 10^{-31}}}$	1
$v = 5.17 \times 10^{5}$ J	1
Total	**6**

The students repeated their original experiment using the caesium (Cs) photocell instead of a scandium (Sc) photocell.

e On the graph for the original experiment, draw a line for the graph that would be obtained for the caesium experiment. (2 marks)

Description	Marks
Cs line drawn parallel to Sc line	1
Cs line drawn to give energy axis intercept of 2.1 eV	1
Total	**2**

Question 20 **(12 marks)**

Quarks are one of the two elementary particles in the Standard Model of particle physics. There are six types of quarks. Each quark has its own antiquark.

Table of quarks			
Name	**Quark symbol**	**Quark electrostatic charge**	**Antiquark symbol**
Up	u	$+\frac{2}{3}e$	$\bar{u}$
Down	d	$-\frac{1}{3}e$	$\bar{d}$
Strange	s	$-\frac{1}{3}e$	$\bar{s}$
Charmed	c	$+\frac{2}{3}e$	$\bar{c}$
Bottom	b	$-\frac{1}{3}e$	$\bar{b}$
Top	t	$+\frac{2}{3}e$	$\bar{t}$

Baryons are made up of three quarks.

The lambda baryon, Λ^0, is composed of an up, down and strange quark. It has a rest mass of 1.98×10^{-27} kg and mean lifetime of 2.63×10^{-10} s.

a Show that Λ^0 baryon is neutral. (1 mark)

Description	Marks
charge = $(+\frac{2}{3}e) + (-\frac{1}{3}e) + (-\frac{1}{3}e) = 0$	1
Total	**1**

b How much energy will be released if a Λ^0 baryon at rest decays to electromagnetic radiation? (2 marks)

Description	Marks
$E = mc^2 = 1.98 \times 10^{-27} \times (3 \times 10^8)^2$	1
$E = 1.78 \times 10^{-10}$ J	1
Total	**2**

c How much energy will be released if a Λ^0 baryon travelling at 0.95*c* decays to electromagnetic radiation? (3 marks)

Description	Marks
$E = \dfrac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} = \dfrac{1.78 \times 10^{-27}}{\sqrt{1 - \frac{(0.95c)^2}{c^2}}}$	1
$E = \dfrac{1.78 \times 10^{-27}}{\sqrt{0.075}}$	1
$E = 5.70 \times 10^{-10}$ J	1
Total	**3**

d What will be the mean lifetime of a Λ^0 baryon when measured in the laboratory if it is travelling at 0.95*c*? (3 marks)

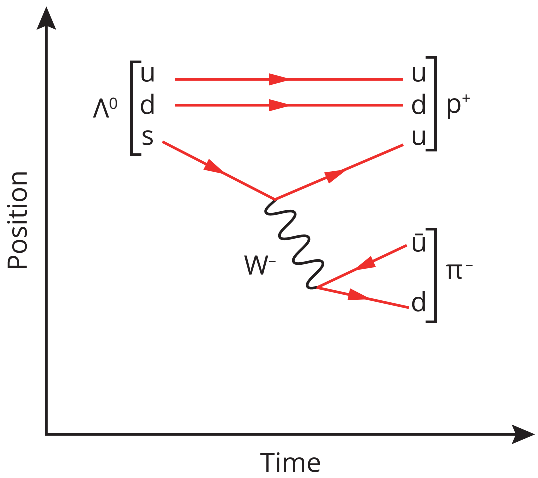
Description	Marks
$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{2.63 \times 10^{-10}}{\sqrt{1 - \frac{(0.95c)^2}{c^2}}}$	1
$t = \frac{2.63 \times 10^{-10}}{\sqrt{0.0975}}$	1
$t = 8.42 \times 10^{-10}$ s	1
Total	**3**

The Λ^0 baryon typically decays to a proton and π^- meson antiparticle, as shown below. This decay is mediated by a W^- boson.

$\Lambda^0 \rightarrow p^+ + \pi^-$

Table of particles for decay of Λ^0	
Particle/antiparticle symbol	**Quark composition**
Λ^0	u d s
p^+	u d u
π^-	d ū

e Complete the Feynman diagram below to represent this decay of the Λ^0 baryon. (3 marks)



Description	Marks
Shows u for p^+, and ū and d for π^-	1
Shows arrows for correct transformations to π^-	1
W^- boson labelled and represented with a wave	1
Total	**3**

Section three: Comprehension (40 marks)

Question 21 **(19 marks)**

Audio induction loops

This symbol is becoming increasingly common in places such as railway platforms, concert theatres, cinemas and seminar rooms. It is the symbol for a piece of technology that assists hearing-impaired people to hear sounds more clearly. The technology is known as an audio induction loop system, also called audio-frequency induction loops or more simply, hearing loops.

Figure 1: Standard symbol to show that a facility has an induction hearing loop installed

A hearing loop consists of a physical wire loop or an array of looped wires placed around an area such as a room or a building. The loop creates a magnetic field around the looped space, which can be picked up by a hearing aid with a telecoil, Cochlear Implant (CI) processors, and specialised hand-held hearing loop receivers for individuals without a telecoil-compatible hearing aid.

The telecoil was originally developed to be fitted into hearing aids to assist hearing-impaired people hear over the telephone. The copper coil picks up electromagnetic fields from coils within a telephone. The telecoil then converts those to an audio signal. Figure 2 shows a representation of a telecoil.

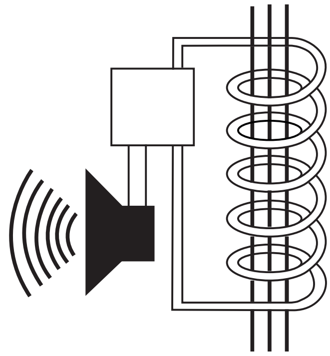


Figure 2: Representation of a telecoil

An audio induction loop system usually consists of a loop of wire around the room (or the space concerned) connected to an amplifier. An audio signal from a microphone, television, stereo or some other source goes to the amplifier, which sends a current through the loop of wire. As the current from the amplifier flows through the loop, it creates a magnetic field within the looped area. This magnetic field is detected by the telecoil, which in turn converts the signal back to an audio signal. In the hearing aid, the telecoil is attached to a loudspeaker that converts the electrical signal to sound. A loudspeaker also works on the principle of induction, using the current to make a diaphragm vibrate, via a magnet, to create sound waves.

The success of an induction hearing loop in a seminar room requires that the speaker's voice be picked up by a microphone. Microphones also work on the principle of induction to convert the sound to an electrical signal and are, in fact, essentially the reverse of a loudspeaker.

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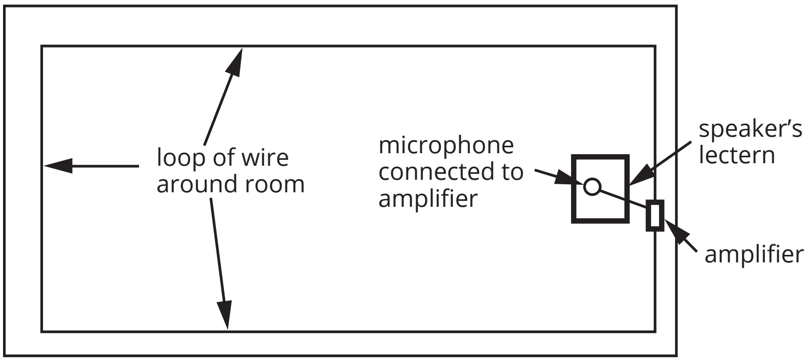
a The rectangle below represents the top view of a seminar room. The front of the room is indicated by a speaker's lectern. Draw a schematic diagram to represent the room fitted with an audio induction loop system. Label the parts of the system required for the loop to operate. (3 marks)



Description	Marks
Loop of wire clearly indicated going around the room	1
Device to transmit sound such as microphone shown connected to amplifier	1
Amplifier shown in circuit	1
Total	**3**

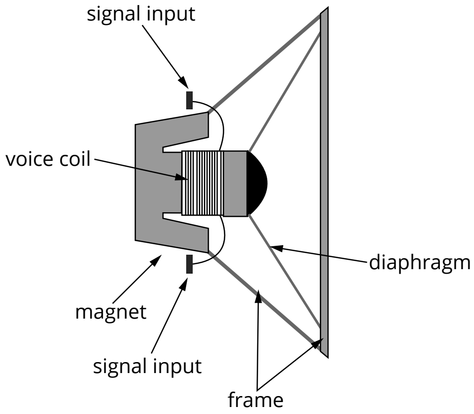
b What type of current is needed for the system to operate? Explain your answer. (2 marks)

Description	Marks
AC is needed.	1
Only AC will induce a magnetic field.	1
Total	**2**

c Identify two ways the size of the induced current in the telecoil may be increased. Give the relationship that supports your answer. (3 marks)

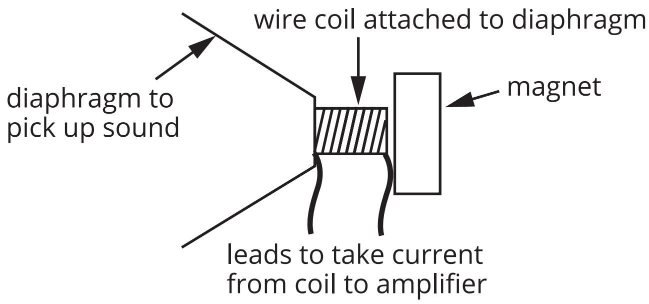
Description	Marks
Increase the number of loops, N, in the coils	1
Increase the strength of the magnet, B, in the coil	1
Induced EMF = $-N\frac{\Delta(BA)}{t}$	1
Total	**3**

d The telecoil converts the signal to sound using a speaker. The diagram below represents a speaker. Explain how a speaker converts the electrical signal to sound. Your answer needs to indicate how the frequency of the sound is produced correctly by the speaker. (5 marks)



Description	Marks
The coil of wire is attached to the speaker cone (diaphragm).	1
When alternating current flows through the wires, it creates a magnetic field.	1
The permanent magnet interacts with the induced magnetic field, producing a force on the coil and diaphragm.	1
The frequency of the AC will be the same as the frequency of the original sound.	1
Thus, the induced magnetic field oscillates at this same frequency so the frequency of the push and pull on the cone resulting from the interaction of the induced magnetic field with permanent magnet will be the same as the original sound.	1
Total	**5**

e Explain how a microphone converts sound to an electrical signal. Support your explanation with a labelled diagram. (6 marks)



Description	Marks
Diagram showing:	
• diaphragm (or similar) to pick up sound	1
• diaphragm attached to a coil that can move	1
• coil near to a permanent magnet	1
Explanation shows recognition that:	
• sound causes the diaphragm to vibrate	1
• the diaphragm causes the coil to vibrate in the magnetic field	1
• alternating current is induced in the vibrating coil	1
Total	**6**

Question 22 **(21 marks)**

Vision

Our capacity to detect light is controlled by four types of light receptors, or photoreceptors. Rods allow us to detect black, white or shades of grey, and the three types of cones are responsible for colour vision. Light is perceived when pigments in the rods and cones capture photons, and it has been shown that they can respond to a single photon.

Rods are able to function at low levels of light and are responsible for night, or scotopic, vision. This sensitivity to light is a result of several rods converging on a single nerve fibre. Rods contain the pigment protein rhodopsin, which undergoes chemical changes on absorption of photons that lead to an electrical signal being sent along the optic nerve to the brain. Rhodopsin absorbs over the wavelengths of approximately 400 to 600 nm, with its maximum absorbance at about 510 nm (Figure 1).

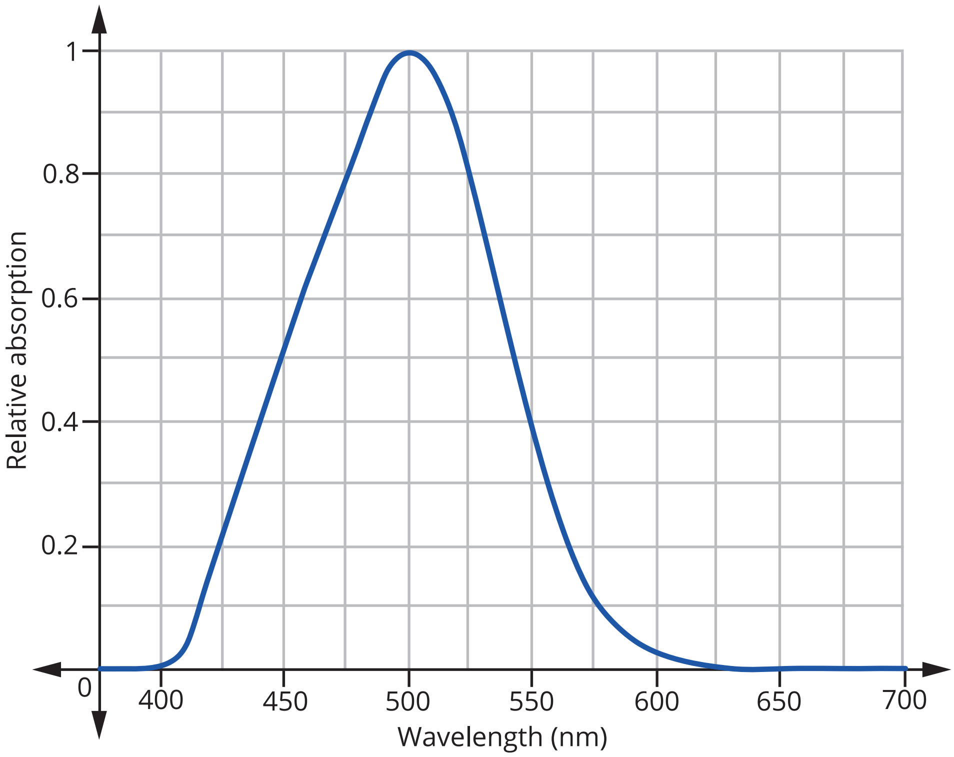


Figure 1: Absorption spectrum for rhodopsin

Humans have three types of cones: L-type cones that are most sensitive to longer wavelength light (red end of the spectrum); M-type cones responding most to medium-wavelength light (green region of the spectrum); and S-type cones detecting short-wavelength light (blue end of the spectrum). The pigments found in cones, opsins, are similar to rhodopsin. On absorbing light, they undergo conformational changes that initiate a series of reactions to produce an electrochemical signal in the optic nerve. The L-type cones are most sensitive and the S-type cones the least sensitive (Figure 2).

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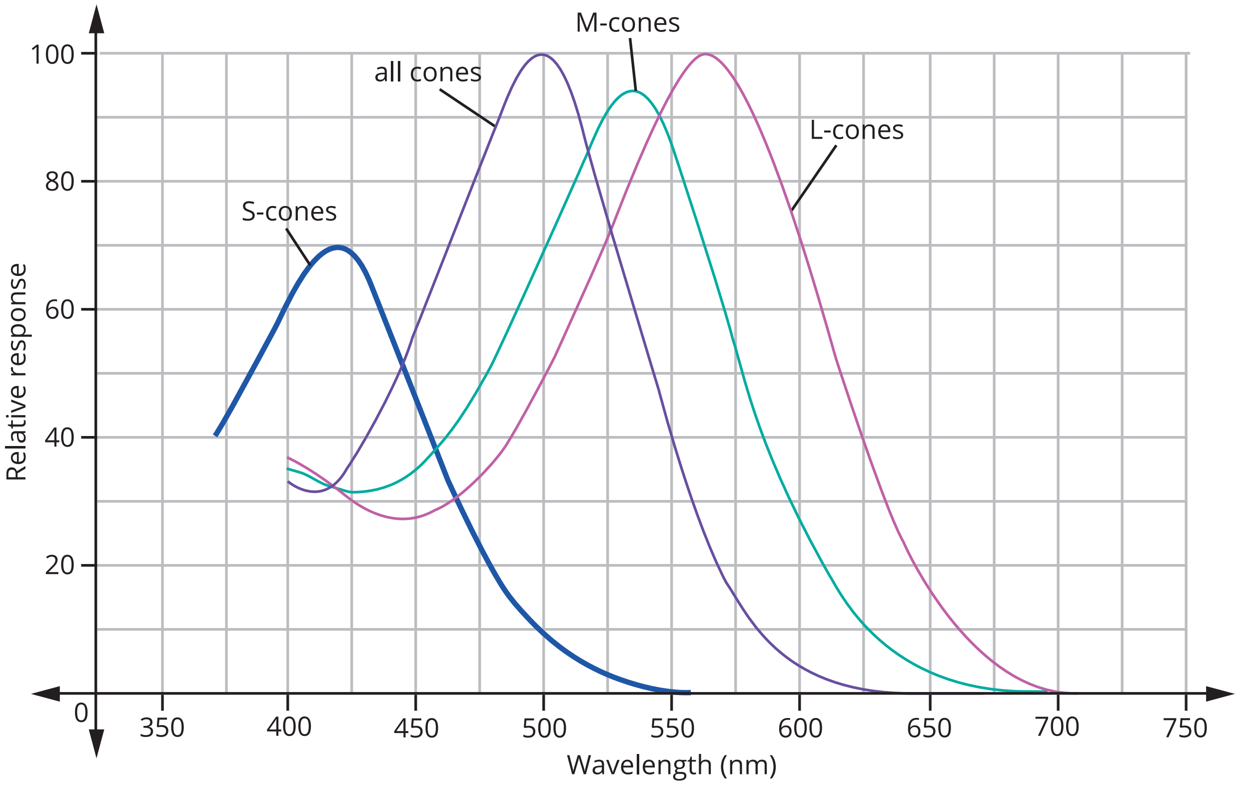

Figure 2: Absorption spectra for cone pigments

Our perception of colour results from the combined responses of cones. If light of wavelength 390 nm enters the eye, only blue light will be perceived. If light wavelength 570 nm enters the eye, M-type and L-type cones will be stimulated and a yellow shade of colour will be perceived.

How bright we perceive light to be depends on the intensity of the light, or the number of photons incident per second. However, not all photons that strike the retina are absorbed by the pigments in the rods and cones—some are transmitted through the retina. In general, the more photons absorbed by the pigments in a given timeframe, the greater the stimulus to the brain. The absorbance of light is wavelength dependant (Figures 1 and 2).

As well as the rods and cones, the lens and cornea have roles in the wavelengths of light which can be sensed by our eyes. The lens does not allow wavelengths shorter than about 380 nm to pass through it, and the cornea is opaque to wavelengths less than 300 nm.

a What is meant by the term 'scotopic vision'? (1 mark)

Description	Marks
Low light or night time vision	1
Total	**1**

b It is stated that rods and cones can respond to a single photon of light. What is the energy associated with a single photon of light of wavelength 550 nm? (2 marks)

Description	Marks
$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{550 \times 10^{-9}}$	1
$E = 3.62 \times 10^{-19}$ J	1
Total	**2**

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c At what frequency are rods most sensitive? To which part of the electromagnetic spectrum does this frequency correspond? (3 marks)

Description	Marks
$c = \lambda f$ Thus, $f = \frac{c}{\lambda} \times \frac{3 \times 10^8}{510 \times 10^{-9}}$	1
$c = 5.88 \times 10^{14}$ Hz	1
This frequency corresponds to about the middle of the visible part of the EMR.	1
Total	**3**

d The article indicates that one factor affecting our perception of brightness is the intensity of the light. What is meant by intensity of light? (1 mark)

Description	Marks
Number of photons incident in a given timeframe	1
Total	**1**

e For light of the same intensity, compare the perception of light of wavelength 510 nm to light of wavelength 460 nm at night. Explain your answer. (4 marks)

Description	Marks
Recognition that no colour is perceived for both wavelengths	1
Because at night the cones are not sensitive enough to respond; only the rods respond	1
Recognition that the 460 nm light will be about half as bright as the 510 nm light	1
Because (taken from Figure 1) rods have twice the absorption at 510 nm as they do at 460 m	1
Total	**4**

f What is the approximate maximum wavelength that can be perceived by human eyes? What colour would this wavelength be perceived as? (2 marks)

Description	Marks
700 nm	1
A shade of red	1
Total	**2**

g What colour would be perceived, in daylight, when light of wavelength 620 nm is incident on the eye? Explain your answer, including an indication of the extent of any response by photoreceptors. (3 marks)

Description	Marks
An orange shade will be perceived.	1
This is due to both M-type and L-type cones getting stimulated.	1
The M-cones have about a 10% response and the L-cones about 50% response.	1
Total	**3**

h If light of wavelength 620 nm with twice the intensity of that used in part f above is incident on the eyes, how will its perception compare to the perception of the light in part f? (2 marks)

Description	Marks
Colour will be the same	1
Will appear twice as bright to the light perceived in part f	1
Total	**2**

i Contrast the way in which the pigments in the rods and cones respond to light to produce an electrical signal to the way that a metal will respond to exposure to light to produce an electrical signal. (3 marks)

Description	Marks
Recognition that the metal will respond by the photoelectric effect	1
Pigments are absorbing photons over a range of wavelengths while a metal will only respond when a particular minimum wavelength of light is reached	1
Pigments produce their electrical signal by an electrochemical process while a metal will produce a current by ejecting electrons from the metal's atoms	1
Total	**3**

Note: Accept any valid comparisons based on the photoelectric effect and the behaviour of eye pigments.

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